An Introduction to 3D Printing

Part 3: Science

Vibrations and Waves, Linear Motion, and Thermal Expansion

Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

Thermal Expansion

Vibrations and Waves

- Tensioned belts connect the toolhead to the rest of the motion system.
- Vibrations are created when the toolhead accelerates.
  - Speeds up.
  - Slows down.
  - Changes direction.
- Belts (strings) + vibrations (waves) = Physics.
- And what does Physics have to say?
  - Motions systems with belts have a dominant **natural frequency**.
  - Think of a guitar string playing a note.

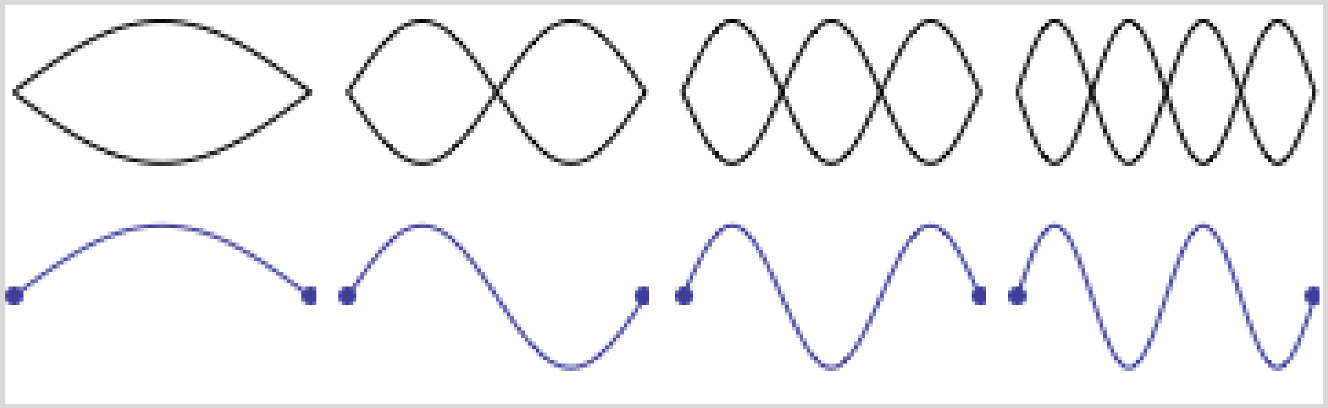
Vibrations and Waves

Let's breakdown what is happening.

- Toolhead excites the belts (adds energy) in the form of vibrations (waves).
- Waves travel along the belts.
  - They interfere and combine with each other. (Superposition principle.)
- In most cases, interference is random.
  - Inconsistent wave patterns.
  - Vibrations stay small and easily dissipate.

Resonance

- A special interference pattern appears at the natural frequency.
  - Standing waves.
- Constructive interference.
- Shape and timing of individual vibrations align and add together.
- Vibrations grow larger and last longer.
- This phenomenon is called resonance.



(Animation by Dan Russell)

Resonance

- Resonance is bad for 3D printing.
- Vibrating belts push and pull on the toolhead → Creates small errors in nozzle position.
- Is there a pattern to these errors?
- **Yes, errors repeat based on the resonant frequency.**
- Imagine writing with pen and paper, and someone bumps your hand one time every second.

- Where do you see evidence of repeating errors?
- Sharp corners → Large accelerations → Large forces (*F* = *ma*).
- Which direction was the toolhead moving?
- From left to right.

(Screenshot from Teaching Tech)

- How can you determine frequency?
- Lookup printing speed in settings.
- Measure distance between repeating patterns.

$$freq = \frac{print\_speed}{repeating\_dist}$$

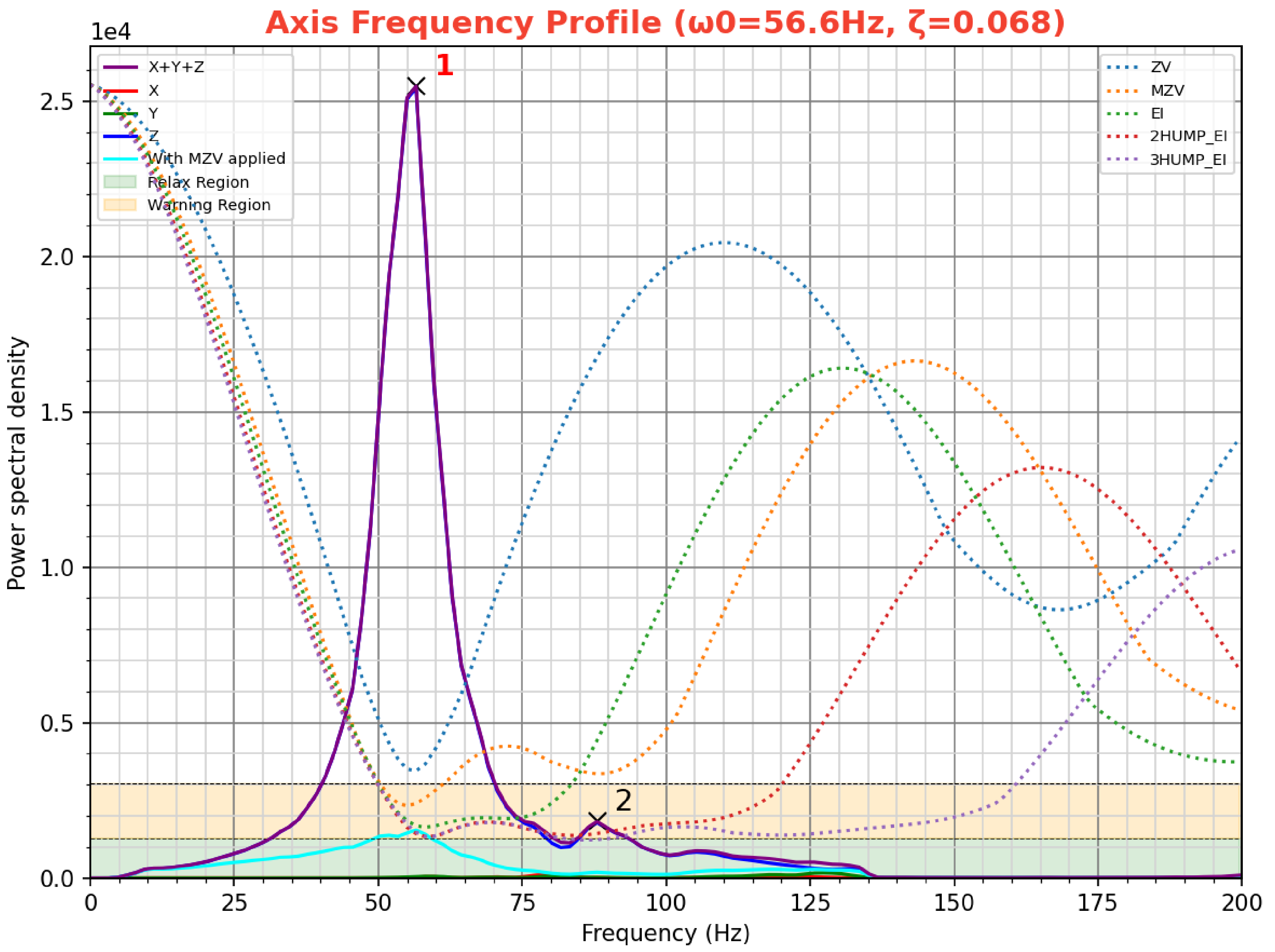
(Screenshot from Teaching Tech)

Input Shaping

- Add extra movements to cancel vibrations.
- Try to create destructive wave interference.
- Opposite of resonance.
- Think noise-canceling headphones.

(Animation, Demo by ACS Motion Control)

- Shake and Tune Plugin.
- Attach an accelerometer to the toolhead and read vibrations.
- Results from my printer.
- **Purple** → Before Input Shaper
  Cyan → After (Estimated)
- Single and narrow peak is ideal because it is easier to fix.
- Typical printing is less than 100 Hz.



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Speed Limits

I want to print faster.

- *Acceleration: 10,000 mm/s²*
- *Max speed: 300 mm/s*
- *Typical extrusions: 0.4 mm wide, 0.2 mm tall*
- *Max volumetric flow: 22 mm³/s*
- **How can I determine what to upgrade?**
  - Motors? Top speed or acceleration?
  - Max flow (heater and nozzle combo)?

Linear Motion

- **What is the distance to reach top speed?**
- Starting speed is 5 mm/s.
  - Cornering speed, square corner velocity, or jerk.
  - *Warning:* Small increases will greatly increase vibrations.
- Need 9 mm extrusions to fully accelerate *and* decelerate.
- Does this seem small enough?

$$v^2 = {v_0}^2 + 2a(x - x_0)$$

$$300^2 = 5^2 + 2 \times 10,000(dist)$$

$$dist = 4.5\ mm$$

Velocity and Flow

- **Can my hotend melt enough plastic at top speed?**
- Almost, 275 < 300.
- Maybe I should upgrade max flow.

$$print\_speed = \frac{flow\_rate}{cross\_section}$$

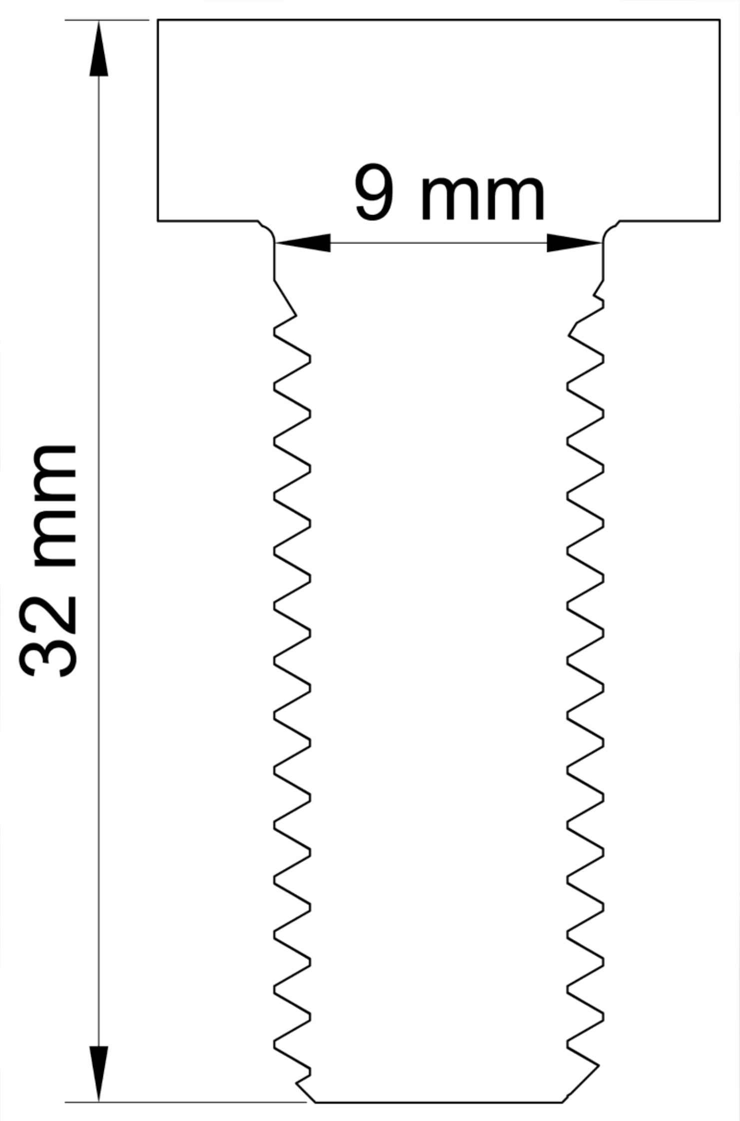
$$print\_speed = \frac{flow\_rate}{extrusion\_width \times layer\_height}$$

$$print\_speed = \frac{22}{0.4 \times 0.2}$$

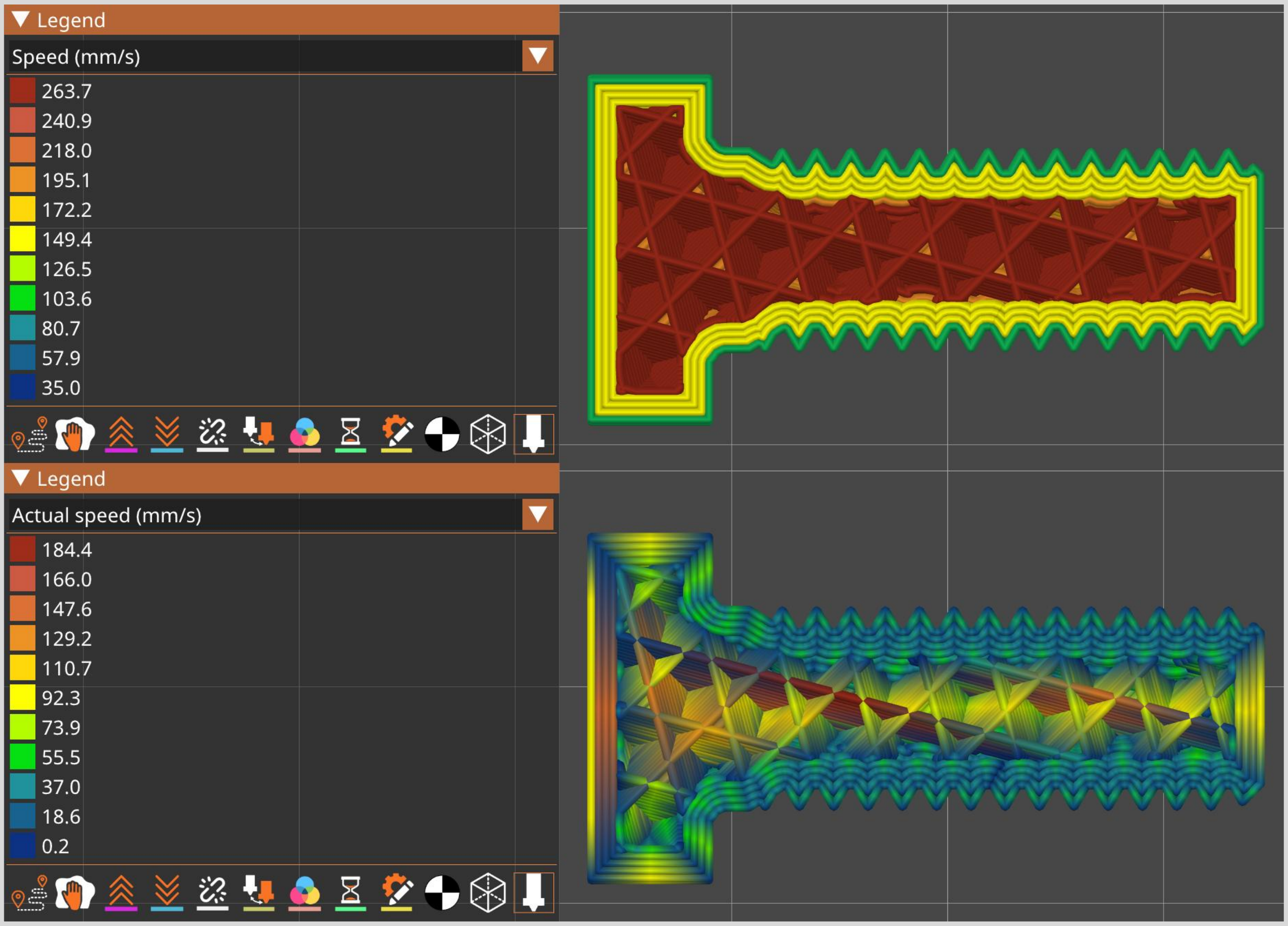
$$print\_speed = 275\ mm/s$$

Finding Limits

- Real life limits are more complex and nuanced.
- *Example:* Printed bolt about an inch long.
- What is going to limit my printer?
- Let PrusaSlicer do all the math!



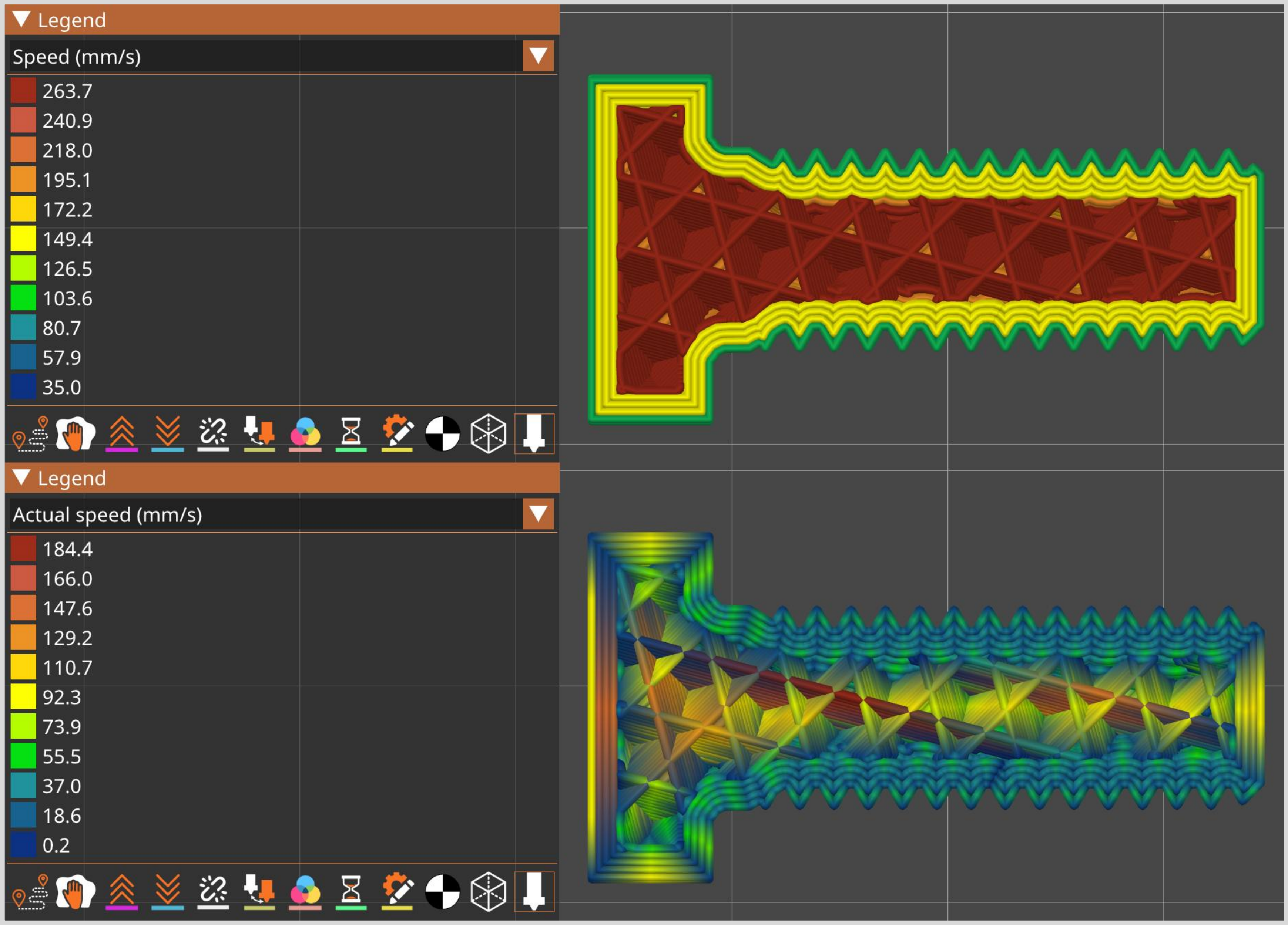
- Max requested speed **264 mm/s**
- Looks close to the estimate.



- Max actual speed **184 mm/s**

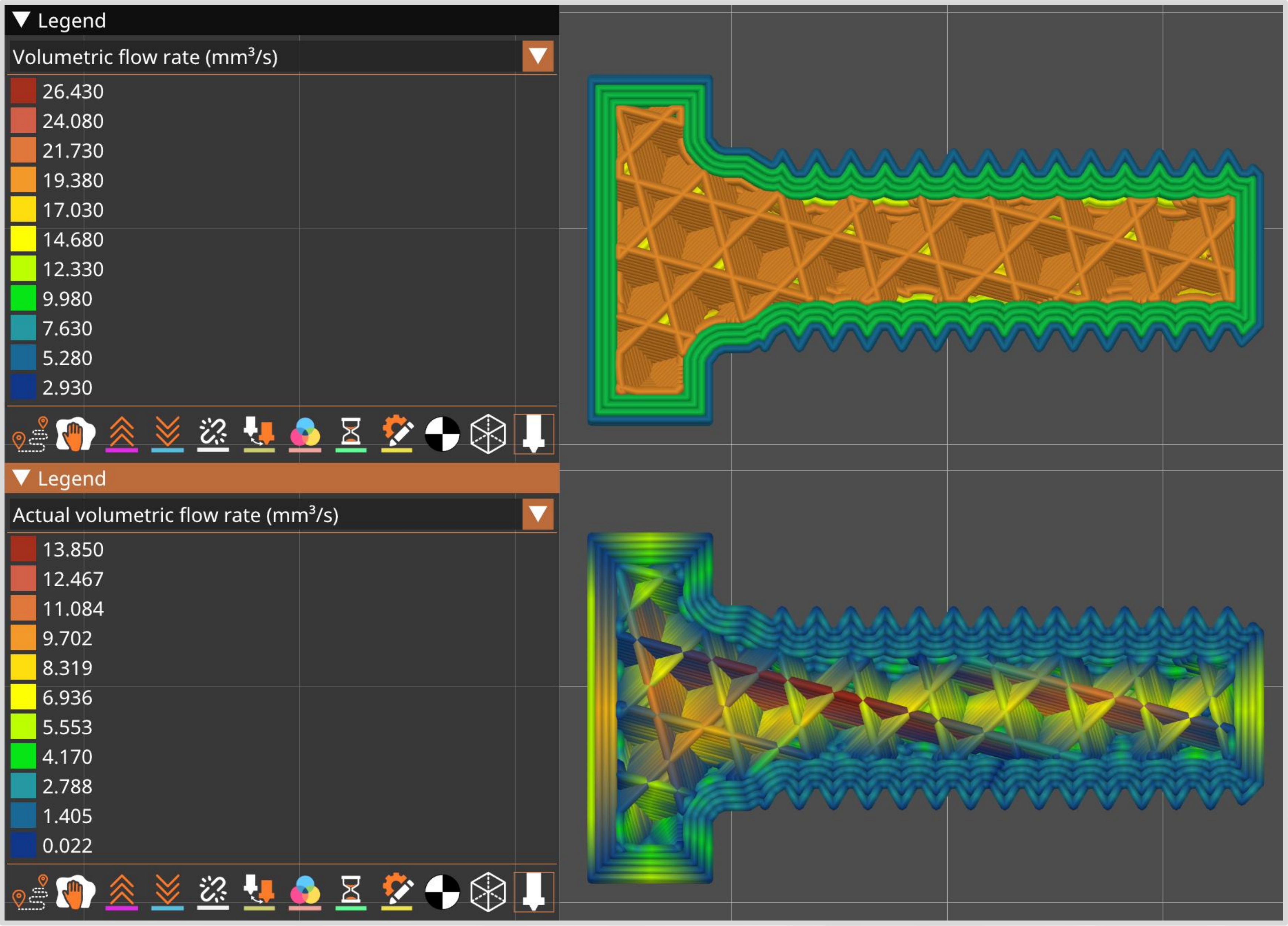
Why is the requested speed non-uniform?

- **Adjust speeds according to visibility**.
- Print slow on the outside to make pretty extrusions.
- Go fast on the inside to save time.



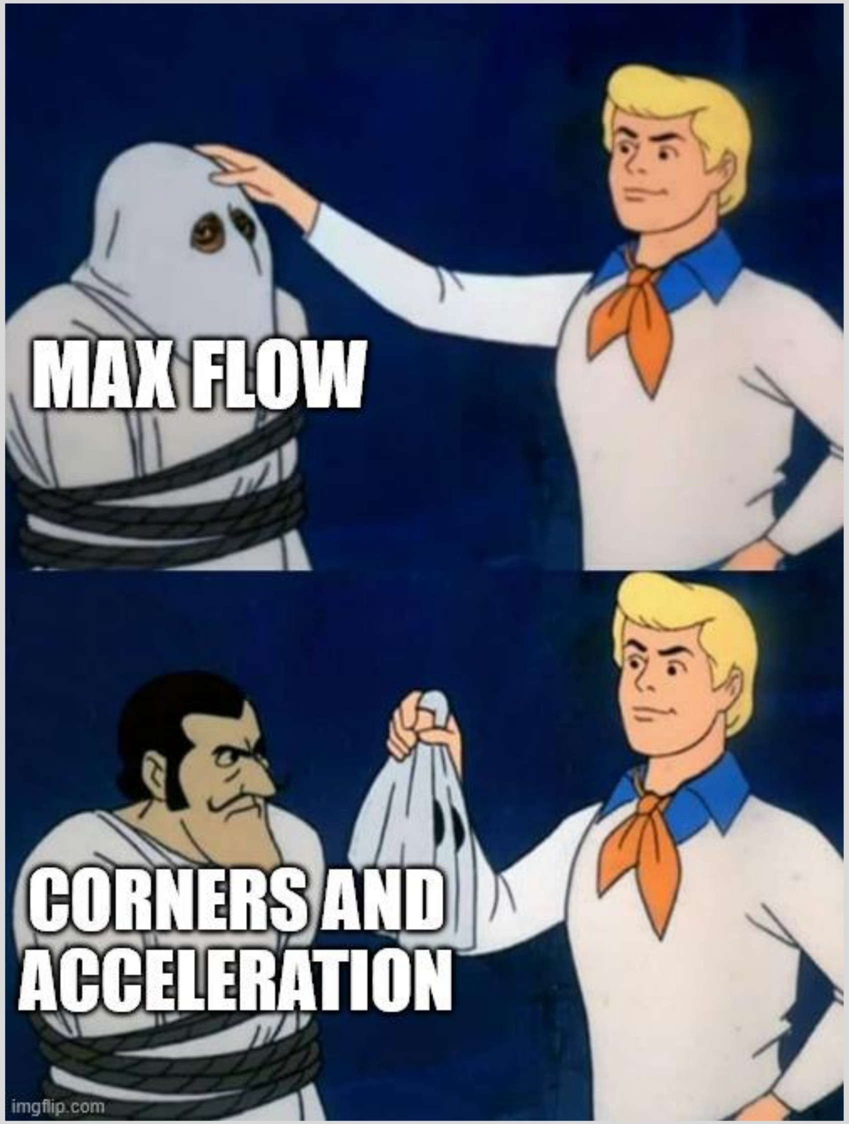
- Max requested flow
  22 mm³/s

- Max actual flow
  14 mm³/s



Speed Limits

- What is going on?
- My printer does not hit max speed, or max flow.
- Limited by corners on the part and acceleration.



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Thermal Expansion

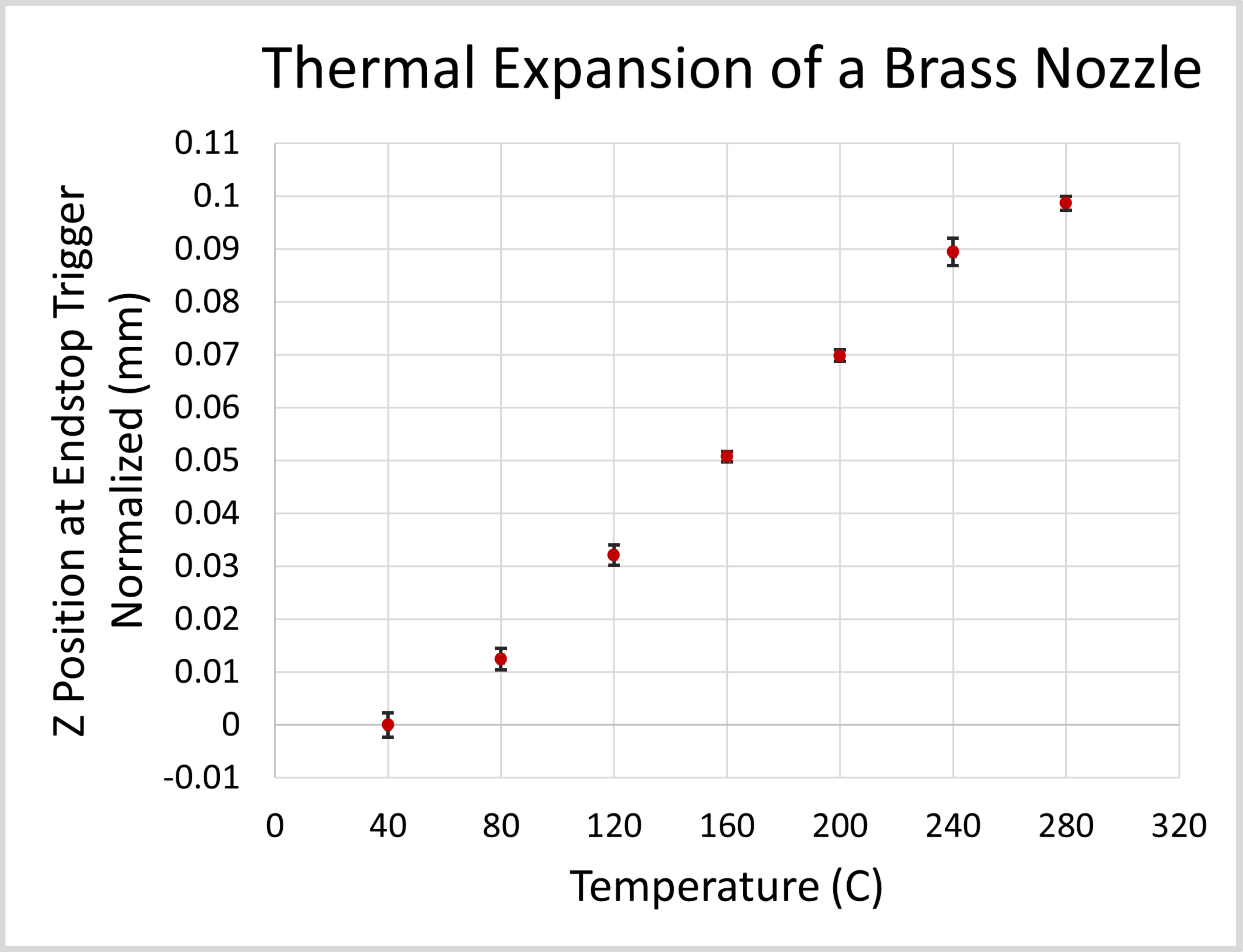
- Many solid materials **expand** as they heat up.
- Intrinsic property called coefficient of linear expansion.
- *Example:* Bridges use expansion joints to prevent gaps and buckling.
    - Grow between 1 and 12 inches
- Is thermal expansion important for tiny 3D printer nozzles?

(Photo by Matt H Wade on Wikipedia.)

Thermal Expansion

- How much longer does a hot brass nozzle grow?
- Simplified estimate with a 12 mm cylinder.
  - ΔT = 150 °C → + 0.034 mm.
  - ΔT = 300 °C → + 0.068 mm.
- Is this enough to be important?
  - Yes, it is large enough to affect the first layer.
  - First layers are **very sensitive.**

- Results from my printer.
- Change temperature then measure nozzle position.
- Does this look like linear expansion?
- Yes, data forms a line.
- My printer uses two probes and calculates a relative offset → Normalized to zero.
- Error bars ± one standard deviation.



Thermal Expansion

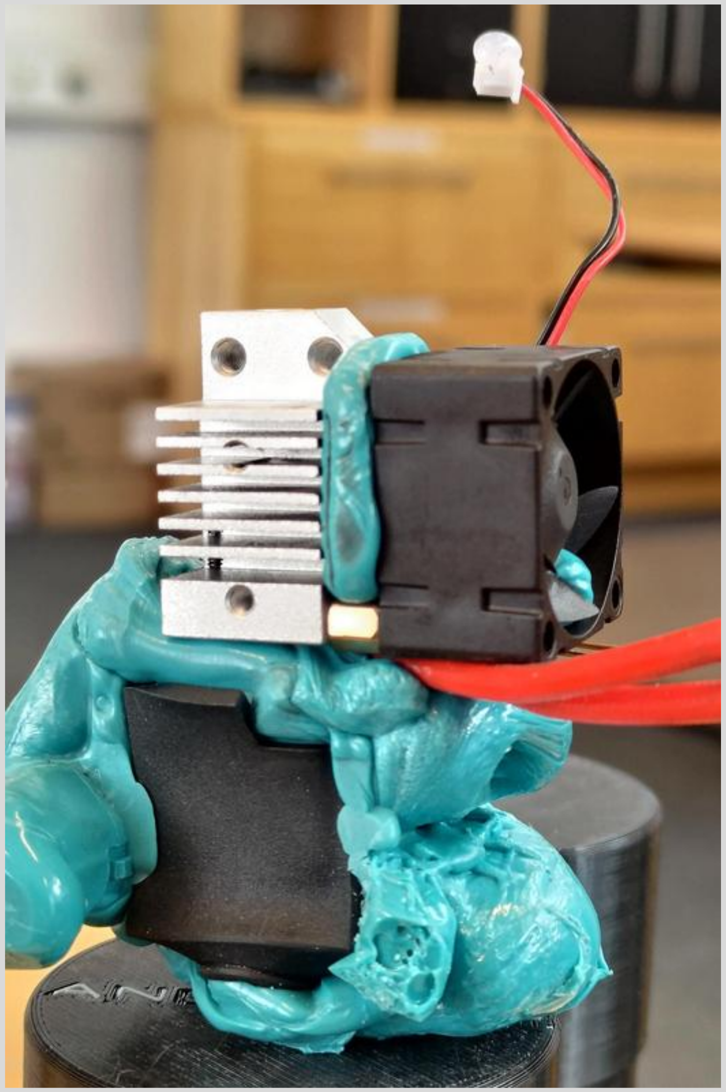
- Extra 0.1 mm at 280 °C is the same as a small layer!
- What can you do about this?
  - "Heat soak." Let everything heat up, expand, and wait for equilibrium.
  - Then probe nozzle position.
  - If you skip waiting, first layer is over-squished.
- Metal print bed needs the same treatment.

Thermal Expansion

- Printers have lots of parts.
- What happens when they expand at different rates?

Hotend Problems

- *Aluminum expands more than brass.*
- Imagine you screw a cold brass nozzle into a cold aluminum heater block.
- Unequal expansion at hot printing temps.
- Aluminum hole expands more than brass nozzle → A tiny gap forms in the threads.
- High pressure from extruding → Filament leaks out.
- Fixed by tightening nozzle at hot printing temps.

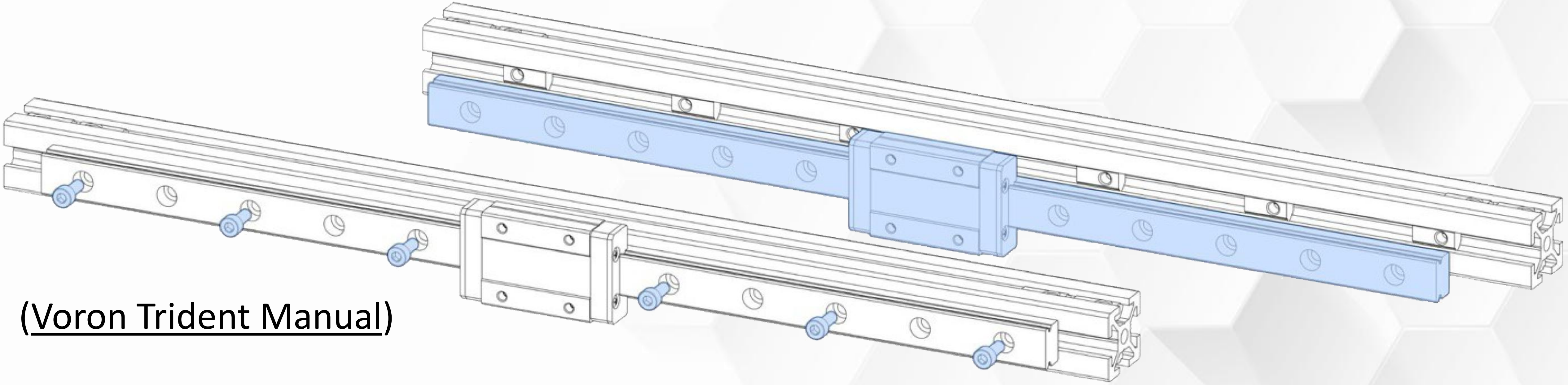


(Blob of doom, reddit post.)

Bimetallic Strips

Consider the X and Y axis on a Voron 3D printer.

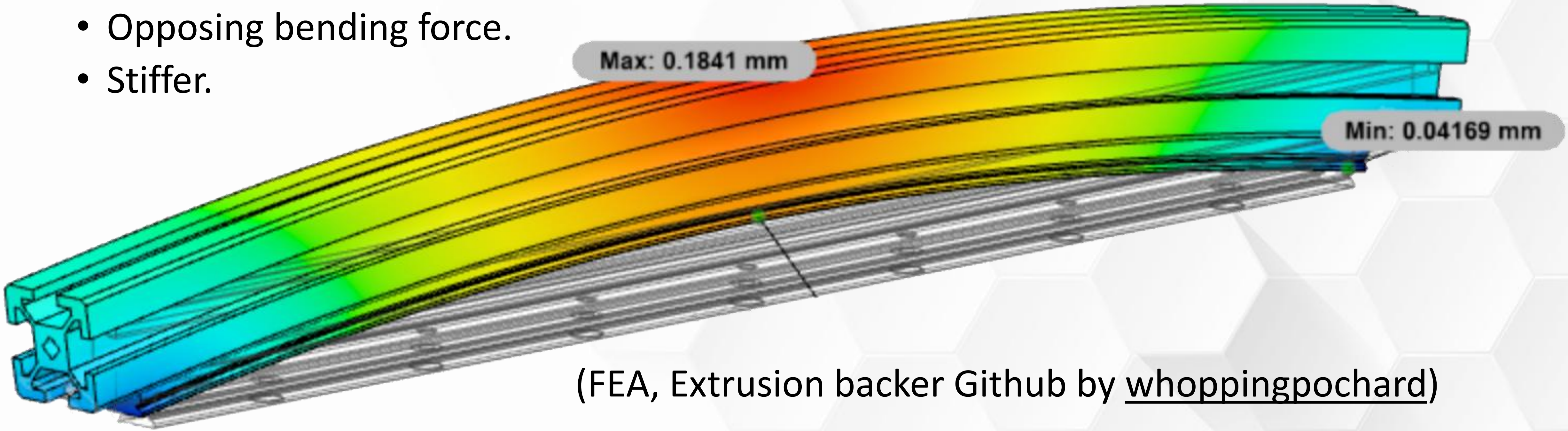
- Aluminum extrusion frame.
- Stainless-steel linear rail.
- Form a bimetallic strip (bar) when screwed together.



(Voron Trident Manual)

Bimetallic Strips

- *Aluminum expands more than stainless-steel.*
- Straight and flat at room temperature → Bows as temperature increases.
- Simulation for 300 mm beam, ΔT = 40°C. Almost 0.2 mm deflection.
- Fixed by adding reinforcing plate.
    - Opposing bending force.
    - Stiffer.



(FEA, Extrusion backer Github by whoppingpochard)

github.com/FlyingGyroscope/IntroTo3DPrinting/